

PIXE Analysis on Maya Blue in Prehispanic and Colonial Mural Paintings

Group_Project_8_Presentation

Awet, Connor, Lachlan, and Pablo

Introduction

Background

- **Maya Blue:** a distinctive blue-green pigment used in Prehispanic and colonial murals across southern Mexico
- **PIXE** (Particle-Induced X-ray Emission): measures elemental concentrations non-destructively
- **Dataset:** 35 mural samples, 17 elements (Na–Pb), from three archaeological groups:
 - *CEC* — Santa Cecilia Acatitlán (post-classic, non-fresco)
 - *Prehispanic* — Cacaxtla, Tajín, Tamuin
 - *Convents* — Mexican Franciscan colonial sites



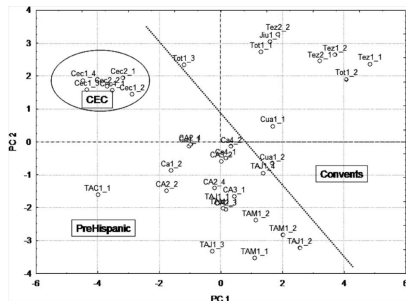
There are **4 main questions** that we set out to answer during this project

- Are there any surprising observations in the data?
- Can we assume that the data follows a multivariate normal distribution?
- Can we find groups in this data?
- Can we recreate figure 2 from the original study?

Figure 2. from the original Paper

- **Original Figure** shows clustering of 3 types of Muarls

- *CEC* Cluster in the top left; Very compact together
- *PreHispanic* Cluster across the bottom; spread through left to middle
- *Convents* Cluster in top right; Cua Moving closer to PreHispanic



- **Original Data** was first cleaned and transformed to remove LOD brackets and change NA's and 0's into usable data
- A **Quantitative** and **Qualitative** analysis was done on the data to compare element concentrations
- **Mahalanobis Distances** were then used to evaluate if certain observations had surprising distances.
- **Multivariate normality** was assessed graphically using a multivariate Q-Q plot of the ordered squared Mahalanobis distances against theoretical chi-squared quantiles, as well as using Mardia MVN test.
- **Principal component analysis** was performed on the covariance matrix of the log-transformed elemental concentrations.
- **Figure 2** from the original paper was recreated using the first two PCs from the Principal component analysis

Original Data

Table 1
Elemental concentration in percent (for the elements marked with asterisk) or $\mu\text{g/g}$ for the samples used in the PCA

	11-Na*	12-Mg*	13-Al*	14-Si*	16-S*	17-Cl*	19-K*	20-Ca*	26-Fe*	15-P	22-Ti	25-Mn	28-Ni	29-Cu	30-Zn	38-Sr	82-Pb
Jiu1_1	0.29	1.76	1.66	10.7	5.23	0.65	0.76	39.3	0.62	[0]	620	(120)	(15)	40	29	440	49
Tot1_1	0.84	1.74	1.85	5.49	2.17	0.35	1.64	35.1	0.61	99,600	840	(170)	(14)	27	164	420	59
Tot1_2	0.20	0.96	1.03	4.55	12.9	0.11	0.12	37.9	0.39	(1200)	810	(130)	(11)	(10)	21	1040	65
Tot1_3	0.45	1.62	2.97	11.6	4.75	0.38	1.71	26.4	1.64	51,800	5010	360	18	35	108	460	69
Tez1_1	0.29	0.68	1.26	5.58	9.89	0.45	0.10	41.6	0.38	[0]	540	(270)	(3)	(8)	(12)	1430	62
Tez1_2	0.32	0.74	1.47	6.57	8.67	0.51	0.15	41.6	0.55	[0]	660	(310)	(6)	(9)	15	1310	152
Tez2_1	1.26	2.79	1.91	7.00	1.83	1.32	0.18	48.8	0.39	[0]	600	(190)	(9)	19	(12)	900	49
Tez2_2	1.71	2.90	2.30	9.77	2.58	1.36	0.26	42.0	0.50	[0]	(520)	310	(14)	31	44	1080	49
Cua1_1	0.36	1.57	2.66	8.32	0.22	(0.03)	0.14	51.7	0.57	[0]	810	330	25	(14)	(14)	8580	21
Cua1_2	0.36	1.68	2.71	8.38	0.18	(0.02)	0.12	51.5	0.51	[0]	820	240	20	(10)	14	6550	1
Ca1_1	0.30	3.17	4.61	17.6	0.39	0.06	0.38	31.8	1.15	(170)	2070	330	26	22	18	560	15
Ca1_2	0.15	3.40	4.12	16.0	0.41	0.05	0.29	34.6	1.41	[0]	1980	630	32	(6)	34	430	2
CA2_1	0.51	3.03	4.52	16.4	0.90	0.08	0.34	33.1	0.99	[0]	1330	600	23	(4)	69	160	9
CA2_2	0.64	2.96	5.80	20.6	0.66	(0.07)	0.59	24.2	1.74	(430)	2340	550	17	(6)	16	100	2
CA2_3	(0.29)	3.19	4.51	16.3	0.43	(0.07)	0.32	34.2	0.75	[230]	2340	(260)	(9)	(2)	8	160	5
CA2_4	0.44	3.22	4.80	17.1	0.47	(0.05)	0.39	32.3	0.76	[270]	2310	(150)	14	(5)	14	210	7
CA3_1	0.49	1.39	3.98	11.6	0.51	0.09	0.29	43.5	0.93	(1300)	2140	(360)	16	(4)	14	160	4
CA3_2	0.70	1.37	3.62	11.7	1.00	(0.07)	0.60	42.2	0.97	(1700)	4560	(320)	(12)	(10)	23	270	7
Ca4_1	0.29	1.64	2.74	10.5	0.84	0.08	0.31	46.1	0.79	(940)	4580	380	22	(7)	25	1180	4
Ca4_2	0.33	1.46	3.08	11.2	0.52	0.07	0.24	45.5	0.78	(1260)	3300	580	18	(11)	26	850	18
Cee1_1	0.23	6.24	3.69	22.8	1.32	0.11	0.71	17.5	2.27	[0]	2110	9120	30	(12)	60	1170	7
Cee1_2	0.11	8.60	2.81	22.5	0.18	0.11	0.58	19.9	1.54	[0]	1820	4790	23	14	48	1260	18
Cee1_3	0.56	9.30	5.29	30.7	0.19	0.06	1.05	2.88	1.32	[0]	1320	1980	21	21	87	490	10
Cee1_4	0.22	8.40	4.54	29.6	0.32	0.06	1.01	6.07	1.54	[0]	1870	6730	26	21	88	810	17
Cee2_1	0.20	6.82	3.50	23.2	1.08	0.11	0.71	17.8	1.67	[0]	1600	8630	29	16	57	1190	13
Cee2_2	0.14	8.51	2.85	23.3	0.65	0.10	0.63	16.6	2.22	[0]	3880	7170	28	16	53	1300	13
TAM1_1	(0.29)	2.25	4.08	12.8	0.15	(0.02)	0.26	41.1	1.49	(850)	1370	(250)	15	(3)	12	500	6
TAM1_2	(0.19)	1.35	3.25	10.5	0.22	[0.01]	0.20	47.3	1.19	(850)	1040	(150)	(12)	(5)	(8)	300	5
TAJ1_1	1.49	0.67	7.43	16.5	(0.16)	(0.07)	0.43	31.7	1.38	[0]	1040	(280)	22	[0]	15	740	10
TAJ1_2	(0.20)	1.26	2.48	8.00	0.10	[0.01]	(0.06)	53.1	0.54	(1530)	850	(170)	(9)	(3)	10	90	3
TAJ1_3	(0.19)	3.37	4.90	17.9	(0.06)	(0.01)	0.27	31.9	0.76	(590)	1470	(140)	12	(5)	8	110	1
TAJ1_4	(0.43)	0.38	2.10	5.10	(0.09)	0.22	0.36	57.8	1.18	(690)	1600	(240)	(29)	(17)	21	570	1
TAC1_1	0.56	1.73	4.90	17.9	0.36	(0.02)	0.90	27.2	5.18	(1000)	3920	900	154	(12)	52	270	1
TAM1_1	0.40	0.98	3.95	10.2	0.14	[0.00]	0.14	47.4	0.86	(1320)	1310	(250)	(11)	(4)	12	320	1
TAM1_2	0.41	0.89	3.32	8.76	0.16	(0.01)	0.14	50.7	0.61	(1340)	1250	(200)	(14)	(3)	(6)	340	4

Numbers in squared brackets are close to the limit of detection (LOD), and numbers in brackets are below the LOD. Typical errors are $\pm 5\text{--}10\%$.

Data Transformation

Transformed Data

Elements	11-Na*	12-Mg*	13-Al*	14-Si*	16-S*	17-Cl*	19-K*	20-Ca*	26-Fe*	15-P	22-Ti	25-Mn	28-Ni	29-Cu	30-Zn	38-Sr	82-Pb
1 ju1_1	-0.53760200	0.245512668	0.22010809	1.0293838	0.71850169	-0.1870866	-0.119186408	1.5943926	-0.207608311	1.929419	2.792392	2.079181	1.1760913	1.6020600	1.4623980	2.643453	1.6901961
2 Tot1_1	-0.07572071	0.240549248	0.26717173	0.7395723	0.336451973	-0.4559320	0.214843848	1.5453071	-0.214670165	4.998259	2.924279	2.230449	1.4313638	2.2148438	2.623249	1.7708520	
3 Tot1_2	-0.68897000	-0.01772867	0.1028372	6.580114	1.11058971	-0.9586073	-0.920818754	1.756392	-0.408935393	3.079181	2.908485	2.113943	1.0413927	1.0000000	1.3222193	3.0167033	1.8129134
4 Tot1_3	-0.209515015	0.47275645	1.0644580	0.67669361	-0.4202164	0.232996110	1.4213690	0.214843848	4.714330	3.698938	2.556303	1.2527325	1.5406680	2.0343238	2.662758	1.8388491	
5 Tez1_1	-0.53760200	-0.167491087	0.10037055	0.7466342	0.99519629	-0.3467875	-1.000000000	1.6190933	-0.420216403	1.929419	2.732394	2.431364	0.4772123	0.9030900	1.0791812	3.155336	1.7923917
6 Tez1_2	-0.49485002	-0.130768280	0.37617173	0.8175654	0.9301910	-0.2924298	-0.823908741	1.6190933	-0.259637311	1.929419	2.819544	2.491362	0.7781513	0.9542425	1.1760913	3.117271	2.1818436
7 Tez2_1	0.10037055	0.445604203	0.28103337	0.8450980	0.26245109	0.1205739	-0.744727495	1.6884198	-0.408935393	1.929419	2.778151	2.278754	0.9542425	1.2787536	1.0791812	2.954243	1.6901961
8 Tez2_2	0.23299611	0.462397998	0.36172784	0.9889496	0.41161971	0.1335389	-0.585026652	1.6232493	-0.301029996	1.929419	2.716003	2.491362	1.1461280	1.4913617	1.6434527	3.033424	1.6901961
9 Cu1_1	-0.44369750	0.195899652	0.42488164	0.9201233	-0.65757732	-1.5228787	-0.853871964	1.7134905	-0.244125144	1.929419	2.908485	2.518514	1.3979400	1.1461280	1.1461280	3.933487	1.3222193
10 Cu1_2	-0.44369750	0.225309282	0.43296929	0.9232440	-0.744727249	-1.6989790	-0.920818754	1.7118072	-0.292429824	1.929419	2.913814	2.380211	1.3010300	1.0000000	1.1461280	3.816241	0.0000000
11 Ca1_1	-0.52287875	0.501059262	0.66370093	1.2455127	-0.40893539	-1.2218487	-0.420216403	1.5024271	0.060697840	2.304449	3.315970	2.518514	1.4149733	1.3424227	1.2552725	2.748188	1.1760913
12 Ca1_2	-0.82390874	0.531478917	0.61489722	1.2041200	-0.38721614	-1.3010300	-0.537602002	1.5390761	0.149219113	1.929419	3.296665	2.799341	1.5051500	0.7781513	1.5314789	2.633468	0.3010300
13 CA2_1	-0.29242982	0.481442629	0.65513843	1.2148438	-0.04575749	-1.0969100	-0.468521083	1.5198280	-0.004364805	1.929419	3.123852	2.778151	1.3617278	0.6020600	1.8388491	2.2041200	0.9542425
14 CA2_2	-0.19382003	0.471291711	0.76342799	1.3138672	-0.18045606	-1.1549020	-0.229147988	1.3838154	0.240549248	2.633468	3.369216	2.740363	1.2304489	0.7781513	1.2041200	2.0000000	0.3010300
15 CA2_3	-0.53760200	0.503790683	0.65417654	1.2121876	-0.36653154	-1.1549020	-0.494850022	1.5340261	-0.124938737	1.929419	3.369216	2.414973	0.9542425	0.3010300	0.9030900	2.2041200	0.6889700
16 CA2_4	-0.35654732	0.507855852	0.68124124	1.2329961	-0.32790214	-1.3010300	-0.408935393	1.5092025	-0.119186408	1.929419	3.361612	2.176091	1.1461280	0.6989700	1.1461280	2.322219	0.8450980
17 CA3_1	-0.30980392	0.143014800	0.59988307	1.0644580	-0.29242982	-1.0457575	-0.537602002	1.6384893	-0.031517051	3.113943	3.330414	2.556303	1.2041200	0.6020600	1.1461280	2.2041200	0.6020600
18 CA3_2	-0.15490196	0.136720567	0.55870857	1.0681859	0.0000000	-1.1549020	-0.221848750	1.6253125	-0.013228266	2.304449	3.658965	2.505150	1.0791812	1.0000000	1.3617278	2.431364	0.8450980
19 Ca4_1	-0.53760200	0.164438804	0.47775056	1.0211893	-0.07572071	-1.0969100	-0.508638306	1.6637909	-0.102372909	2.973128	3.680685	2.579784	1.3424227	0.8450980	1.3979400	3.071882	0.6020600
20 Ca4_2	-0.48148606	0.248432856	0.48855072	1.0492180	-0.2839966	-1.1549020	-0.619788756	1.6580114	-0.107995397	3.100371	3.518514	2.763428	1.2552725	1.0413927	1.449733	2.929419	1.2552725
21 Cc1_1	-0.63827216	0.795184590	0.56702637	1.5793948	0.12057393	-0.9586073	-0.148741651	1.2430380	0.356025857	1.929419	3.324282	3.959995	1.4771213	1.0791812	1.1761513	3.068186	0.8450980
22 Cc1_2	-0.95860731	0.93449881	0.44873632	1.3521825	-0.744727249	-0.9586073	-0.236572006	1.2988531	0.187520721	1.929419	3.260071	3.680336	1.3617278	1.1461280	1.6812412	1.3100371	1.2552725
23 Cc1_3	-0.25181197	0.968482949	0.72345567	1.4871384	-0.72124640	-1.2218487	-0.021189299	0.4593925	0.120573931	1.929419	1.320574	3.96665	1.3222193	1.9935193	2.690196	1.0000000	
24 Cc1_4	-0.65757732	0.92479286	0.60575585	1.4712917	-0.49485002	-1.2218487	0.004321374	0.7831887	0.187520721	1.929419	3.271842	3.828015	1.4149733	1.3222193	1.9444827	2.908485	1.2304489
25 Tez2_1	-0.68897000	0.833784375	0.54406804	1.3654880	0.03342376	-0.9586073	-0.148741651	1.2504200	0.222716471	1.929419	2.304120	3.96011	1.4623980	1.2041200	1.7558749	2.755747	1.1139434
26 Cc2_2	-0.85387196	0.929299560	0.54544486	1.3673559	-0.18708664	-1.0000000	-0.200659451	1.2201081	0.346352974	1.929419	3.588832	3.855519	1.4471580	1.2041200	1.7242759	3.113943	1.1139434
27 TAM1_1	-0.53760200	0.35218528	0.61066016	1.1072100	-0.82390874	-1.6989700	-0.585026652	1.6138418	0.173186268	2.929419	3.136721	2.397940	1.1760913	0.4772123	1.0791812	2.698970	0.7781513
28 TAM1_2	-0.72124640	0.310333768	0.51188336	1.0211893	-0.65757732	-2.3010300	-0.698970004	1.6748611	0.075546961	2.929419	3.017033	2.176091	1.0791812	0.6989700	0.9030900	2.471221	0.6989700
29 TAJ1_1	-0.71318627	-0.173925197	0.87098881	1.2174839	-0.79584002	-1.1549020	-0.366531544	1.5010593	0.139879086	1.929419	3.017033	2.447158	1.3424227	0.0000000	1.1760913	2.869232	1.0000000
30 TAJ1_2	-0.68897000	0.100370545	0.39455168	0.9030900	-1.0000000	-2.3010300	-1.221848750	1.7250595	-0.267606240	3.184691	2.929419	2.518514	0.9542425	0.4772123	1.0000000	1.954243	0.4772123
31 TAJ1_3	-0.72124640	0.527629901	0.69019608	1.2528530	-1.22184875	-2.0000000	-0.58636236	1.5037907	-0.119186408	2.770852	1.167317	2.146128	1.0791812	0.6989700	1.3020993	2.041393	0.0000000
32 TAJ1_4	-0.36653154	-0.420216403	0.32221929	0.7075072	-1.04575749	-0.5575773	-0.443697499	1.7619278	0.077188207	2.838849	2.304120	3.862128	1.4623980	1.2304489	0.9232193	2.755875	0.0000000
33 TACL1_1	-0.25181197	0.238061013	0.69019608	1.2528530	-0.44369750	-1.6989700	-0.045757491	1.4345689	0.714329760	3.000000	3.593286	2.95243	2.1875207	1.0791812	1.1760033	2.431364	0.0000000
34 TAM1_1	-0.39794001	-0.008773924	0.59659710	1.0086002	-0.85387196	-2.3010300	-0.853871964	1.6757783	-0.065501549	3.120574	3.117271	2.397940	1.0413927	0.6020600	1.0791812	2.505150	0.0000000
35 TAM1_2	-0.38721614	-0.050609993	0.52213808	0.9425041	-0.79584002	-2.0000000	-0.853871964	1.7050080	-0.214670165	3.127105	3.096910	2.301030	1.1461280	0.4772123	0.7781513	2.531479	0.6020600

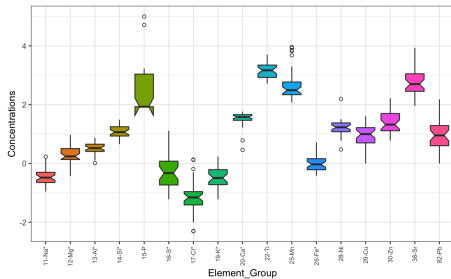
Descriptive Statistics

Descriptive Statistics of Elemental Concentrations

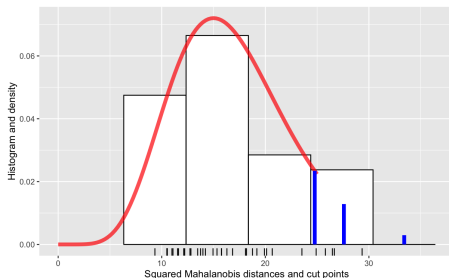
Log10 Transformation

Element	Mean	Standard D.	Median	Minimum	1st Quartile	Inter-Q Range	3rd Quartile	Maximum
11-Na*	-0.44	0.28	-0.48	-0.96	-0.65	0.35	-0.30	0.23
12-Mg*	0.33	0.35	0.24	-0.42	0.13	0.37	0.51	0.97
13-Al*	0.50	0.19	0.52	0.01	0.41	0.24	0.65	0.87
14-Si*	1.09	0.22	1.06	0.66	0.93	0.32	1.25	1.49
16-S*	-0.23	0.61	-0.33	-1.22	-0.73	0.81	0.08	1.11
17-Cl*	-1.14	0.63	-1.15	-2.30	-1.41	0.45	-0.96	0.13
19-K*	-0.47	0.35	-0.49	-1.22	-0.72	0.51	-0.21	0.23
20-Ca*	1.50	0.26	1.58	0.46	1.47	0.19	1.66	1.76
26-Fe*	-0.02	0.25	-0.03	-0.42	-0.21	0.38	0.16	0.71
15-P	2.53	0.78	1.93	1.93	1.93	1.11	3.04	5.00
22-Ti	3.18	0.28	3.17	2.72	2.93	0.42	3.35	3.70
25-Mn	2.67	0.55	2.49	2.08	2.34	0.43	2.77	3.96
28-Ni	1.23	0.27	1.23	0.48	1.08	0.30	1.38	2.19
29-Cu	0.95	0.38	1.00	0.00	0.70	0.52	1.22	1.60
30-Zn	1.38	0.37	1.32	0.78	1.11	0.59	1.70	2.21
38-Sr	2.74	0.45	2.70	1.95	2.45	0.60	3.05	3.93
82-Pb	0.96	0.61	0.95	0.00	0.60	0.69	1.29	2.18

Qualitative Analysis



Are There Surprising Observations?



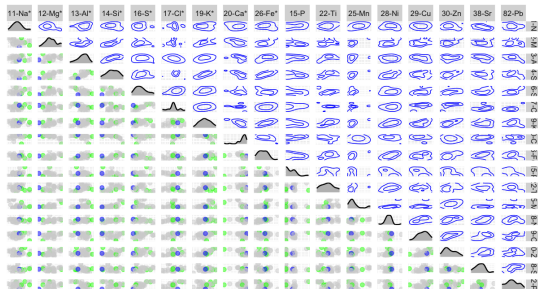
- The 90%, 95%, and 99% Cut off values were: 24.76904, 27.58711, and 33.40866 respectively
- We Identified 1 surprising and 4 somewhat surprising observations

Closer Look at Suprising Observations

Sample	dM	surprise
TAJ1_4	29.34327	Suprising
Tot1_1	26.66402	somewhat
TAJ1_1	26.47125	somewhat
Jiu1_1	25.79784	somewhat
Cec1_3	24.90653	somewhat

- The largest Mahalanobis distances identify samples with unusual combinations of elemental concentrations.
- TAJ1_4 is the most extreme observation, suggesting that its overall elemental profile differs most strongly from the center of the dataset.

Pairs Plot of Mahalanobis Distances and Contour Densities



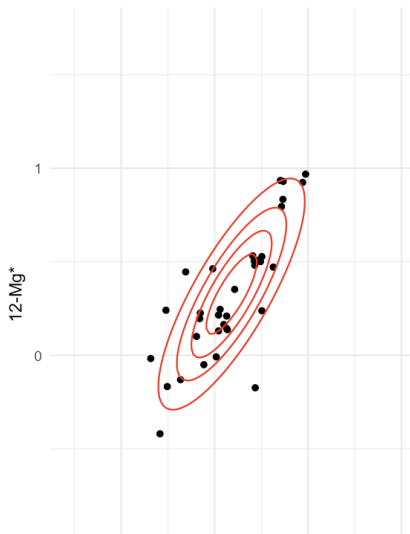
Mahalanobis classification

● Typical (<90%) ● Somewhat surprising (90–95%) ● Surprising (95–99%) ● Very surprising (>99%)

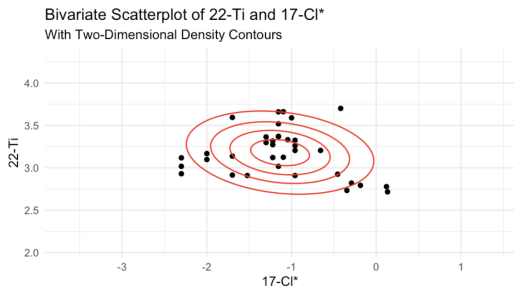
- Each panel compares two elements pairwise
- Contours show where observations are most concentrated
- Highlighted points flag unusual multivariate behaviour
- Outliers are rarely extreme in a single element — their unusualness is multivariate

Contour plot 1

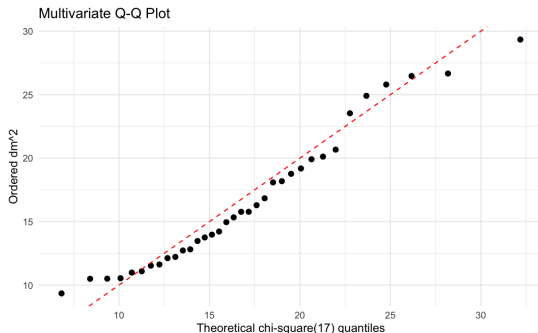
Bivariate Scatterplot of $^{12}\text{Mg}^*$ and $^{14}\text{Si}^*$
With Two-Dimensional Density Contours



Contour plot 2



Can we assume that the data follows a multivariate normal distribution?



- Fails Mardia's skewness test ($p = 0.003$) — **not multivariate normal**
- Kurtosis result is consistent with normality

Elements that aren't normally distributed

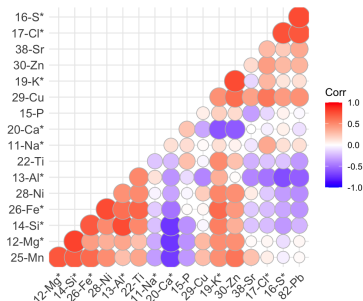
Anderson-Darling Tests were conducted on each element to check individual normality

H0: Normally Distributed H1: Not Normally Distributed $\sim \alpha = 0.05$

- Anderson-Darling 20-Ca* 2.670 < 0.001 Not normal
- Anderson-Darling 15-P 3.143 < 0.001 Not normal
- Anderson-Darling 25-Mn 2.809 < 0.001 Not normal
- Anderson-Darling 28-Ni 0.812 0.032 Not normal

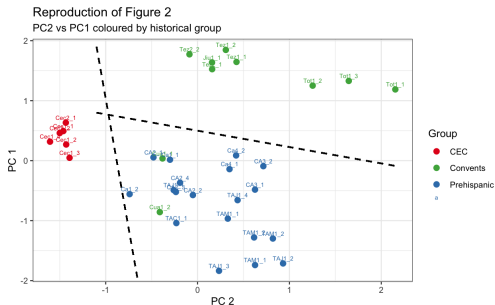
Note: 15-P more than likely Not normal due to data transformation due to large number of missing values.

Correlogram



- Red = positive correlation, blue = negative; larger circles indicate stronger relationships
- Mn, Mg, Si, Fe, Ni, Al, Ti, K, and Zn are mutually positively correlated
- Na, Ca, P, S, and Pb are negatively or uncorrelated with that group

Can we find groups in this data?



- **CEC** forms a tight, well-separated cluster
- **Prehispanic** and **Convents** overlap substantially
- Some structure exists, but groups are not fully distinct

The data also showed some surprising observations. TAJ1_4 had the biggest Mahalanobis distance and was classified as surprising, while Tot1_1, TAJ1_1, Jiu1_1 and Cec1_3 were all classified as somewhat surprising. The clustering analysis provides evidence of some meaningful groups in the data, with the CEC samples forming a distinct group. This is consistent with the authors' observation of these samples containing unusually low calcium concentrations and likely not being produced with the fresco technique. The samples also show a less clear distinction between predominantly Prehispanic and Colonial samples, as these two groups have substantial overlap rather than being completely separated.

We agree with the authors' conclusions about the zones in Figure 2. The original PCA differentiated samples into broad Prehispanic and Colonial patterns and identified CEC as a distinct cluster. They also concluded that the data did not form well-defined groups. We were able to reproduce a similar graph, providing evidence of these findings. Elemental composition provides information about the historical period and the technique used, but the overlap between samples means that the groups cannot be treated as completely separated categories. The authors' hypothesise that the overlap may be due to similarities in the painting techniques between the Prehispanic and Colonial periods.

Conclusion

Overall, the results show some structure in the Maya blue samples, but the groups are not completely distinct. The CEC samples are clearly separated, while the Prehispanic and Colonial samples overlap. Figure 2 supports the authors' conclusion that the zones represent broad patterns rather than clearly defined groups.

Sánchez del Río, M., Martinetto, P., Solís, C., & Reyes-Valerio, C. (2006). *PIXE analysis on Maya blue in Prehispanic and colonial mural paintings*. Nuclear Instruments and Methods in Physics Research Section B, 249, 628–632.